

MEESHO PRAGATI AGENT

Agentic AI for Fair, Explainable Credit Access

Project Technical Diagrams Report

Theme: Building for Bharat with the Power of Agentic AI

Target Vertical: Meesho Finance (MPPL) & NBFC Lending Partners

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1. Complete Workflow Diagram



Description

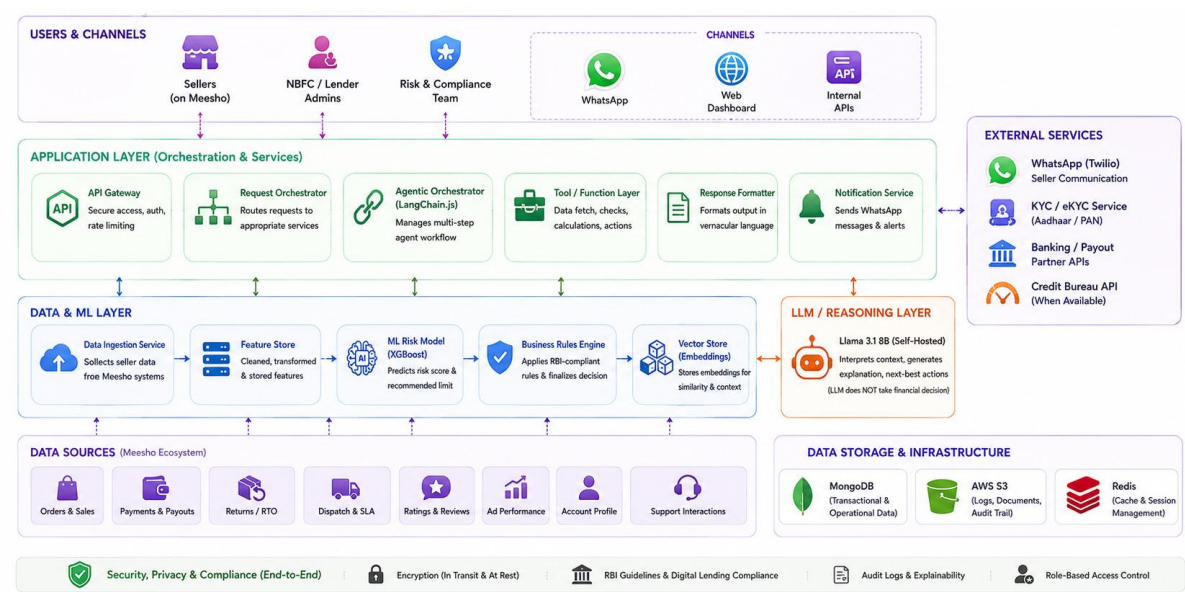
The Complete Workflow Diagram illustrates the end-to-end operational flow of the Meesho Pragati Agent, beginning from the moment a seller initiates a loan request and ending with post-loan monitoring and continuous engagement. The workflow demonstrates how multiple AI, Machine Learning, business rule, and communication components collaborate to generate transparent, explainable, and RBI-compliant lending decisions.

Initially, the system collects the seller's historical performance data, including sales trends, return rates, dispatch SLA, customer ratings, advertising performance, repayment history, and account activity from the Meesho ecosystem. This data undergoes preprocessing and feature engineering before being passed to the Machine Learning model for risk prediction.

The trained XGBoost model generates a numerical risk score along with a recommended loan limit. These outputs are validated through a deterministic Business Rules Engine that applies lending policies, exposure limits, fraud detection, eligibility criteria, and RBI compliance rules.

Once the financial decision is finalized, the Agentic AI layer interprets the model's outputs and generates human-readable explanations in the seller's preferred language. The decision, explanation, coaching tips, and repayment information are then delivered via WhatsApp or the seller dashboard. Finally, the system continuously monitors seller performance and repayment behavior, allowing periodic re-evaluation and personalized financial coaching to support long-term business growth.

2. High-Level System Architecture Diagram



Description

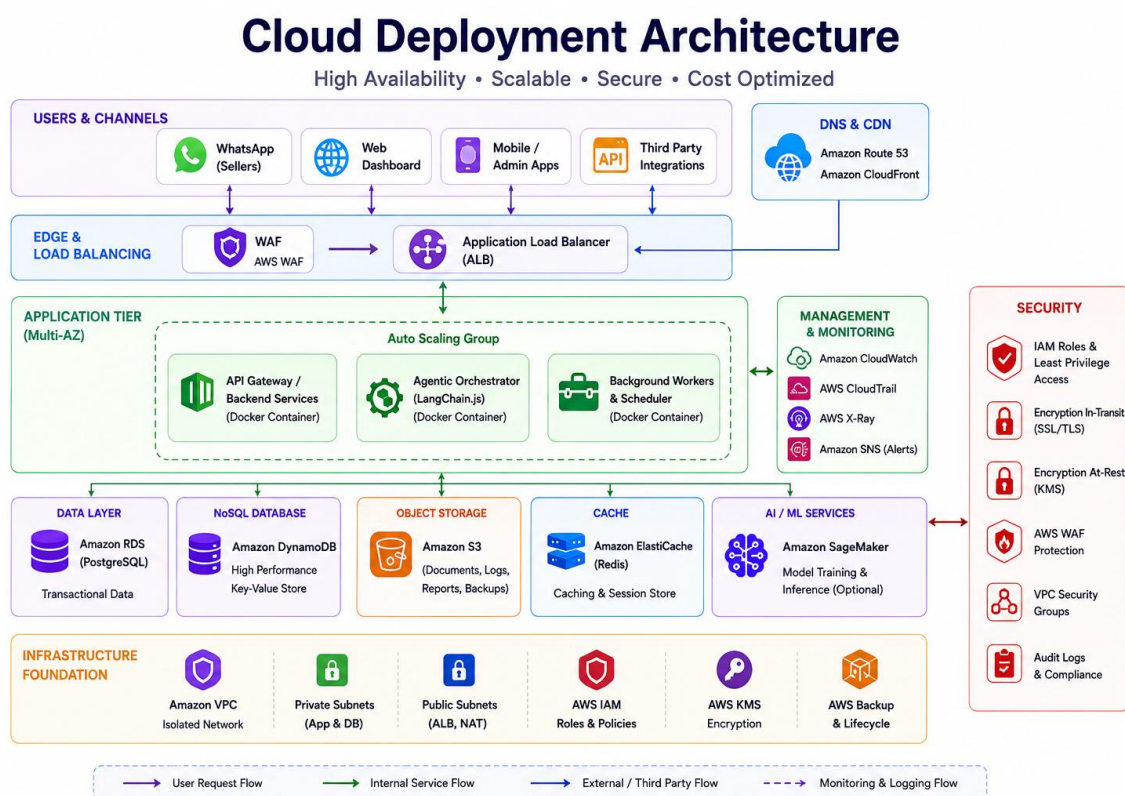
The High-Level System Architecture Diagram presents the major software components of the Meesho Pragati Agent and illustrates how they interact to deliver an intelligent lending solution. The architecture is organized into multiple logical layers that separate user interaction, application orchestration, AI processing, data management, and infrastructure.

Users interact with the system through WhatsApp, web dashboards, mobile applications, or internal APIs. All requests are received by the API Gateway and forwarded to the central Agentic Orchestrator developed using Node.js, Express.js, and LangChain.

The orchestrator coordinates communication between various backend services, including data ingestion, feature engineering, machine learning inference, business rule evaluation, LLM reasoning, and communication services. The AI layer combines deterministic Machine Learning models with Large Language Models to ensure both accurate predictions and explainable decisions.

Persistent storage is handled through MongoDB, AWS S3, Redis, and feature stores, while deployment, monitoring, authentication, and security services operate independently in the infrastructure layer. This modular architecture ensures scalability, maintainability, fault isolation, and secure service integration.

3. Cloud Deployment Architecture



Description

The Cloud Deployment Architecture Diagram illustrates how the Meesho Pragati Agent is deployed within a secure and scalable cloud environment using AWS services. It demonstrates

Monitoring services including CloudWatch, CloudTrail, and X-Ray continuously track application performance, security events, and operational metrics. Security is enforced through IAM roles, encrypted communication, VPC isolation, TLS, KMS encryption, and audit logging. The architecture supports horizontal scaling through Auto Scaling Groups and ensures high availability using Multi-AZ deployment.

Description

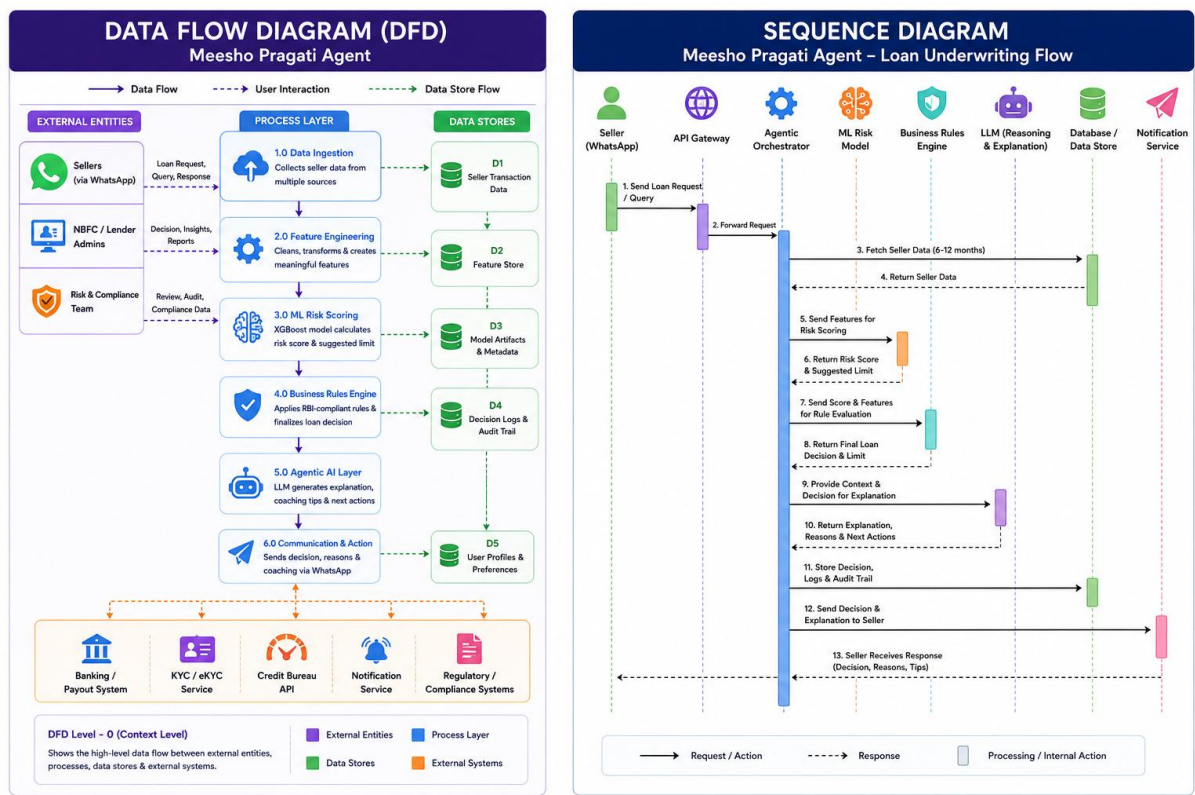
The Data Flow Diagram (DFD) illustrates the movement of information throughout the Meesho Pragati Agent ecosystem. It identifies all external entities, processing modules, data stores, and system interactions involved during loan underwriting.

External entities such as sellers, NBFC administrators, compliance teams, and third-party financial services exchange information with the system. Incoming seller data is first processed by the Data Ingestion Service before moving into the Feature Engineering module, where raw business information is transformed into structured Machine Learning features.

The processed features are stored within dedicated feature repositories and supplied to the Machine Learning Risk Scoring Service. The predicted outputs are subsequently evaluated by the Business Rules Engine before being forwarded to the Agentic AI reasoning layer for explanation generation.

The finalized decision, audit logs, feature data, and communication history are securely stored in multiple databases while notifications are transmitted back to users through WhatsApp and dashboard interfaces. The DFD highlights how information flows across different components while maintaining data consistency, traceability, and security.

5. Sequence Diagram



Description

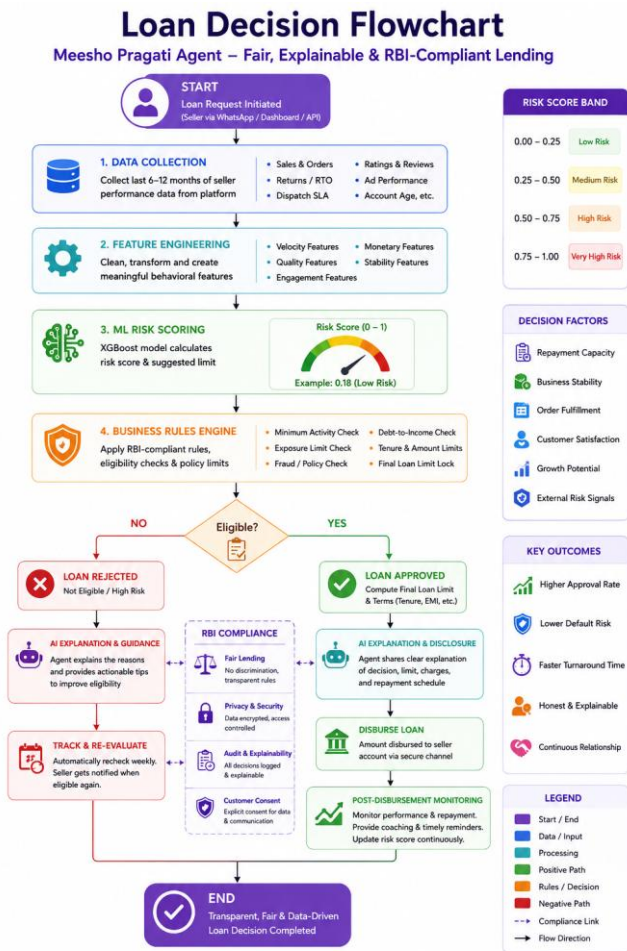
The Sequence Diagram represents the chronological interaction among various system components during the loan evaluation process. It demonstrates how requests propagate through different services and how responses are generated step by step.

The process begins when a seller submits a loan request through WhatsApp or the web dashboard. The API Gateway authenticates the request and forwards it to the Agentic Orchestrator, which retrieves seller information from the database.

After data retrieval, the orchestrator invokes the Feature Engineering Service followed by the Machine Learning Risk Scoring Service. Once the model predicts the seller's risk score and recommended loan amount, the Business Rules Engine validates these results against lending policies and compliance requirements.

The validated outputs are forwarded to the LLM Reasoning Service, which generates transparent explanations and coaching messages. Finally, the decision, explanation, repayment schedule, and audit logs are stored, and the seller receives the response through the communication service. The sequence diagram clearly represents service dependencies, execution order, and synchronous communication among components.

6. Loan Decision Flowchart



Description

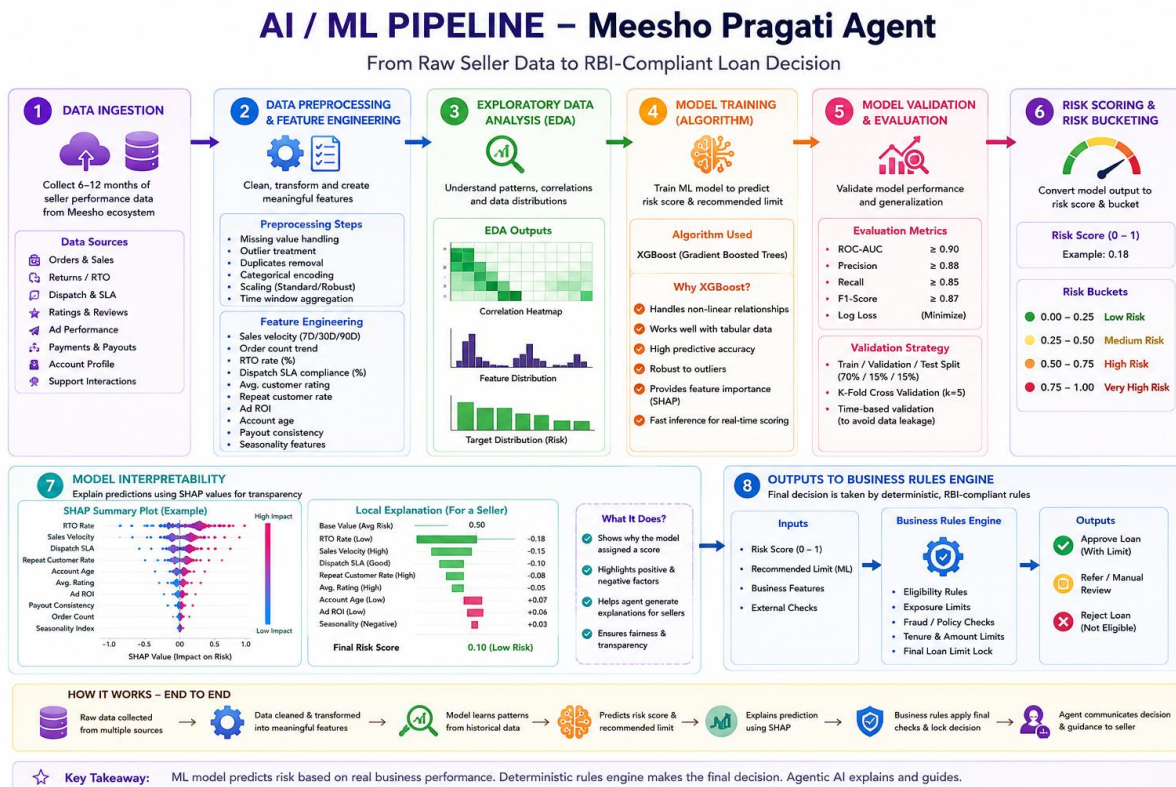
The Loan Decision Flowchart describes the complete decision-making process followed by the Meesho Pragati Agent while evaluating loan applications. It visually represents every decision node, eligibility criterion, validation step, and possible outcome within the lending workflow.

The process begins with seller data collection, where business performance metrics are gathered from the Meesho ecosystem. After preprocessing and feature engineering, the Machine Learning model calculates a numerical risk score representing the probability of repayment.

The Business Rules Engine then applies predefined lending policies, fraud detection rules, exposure limits, account activity requirements, and RBI compliance checks. Based on these validations, the system determines whether the applicant qualifies for loan approval.

Eligible sellers receive a finalized loan amount, repayment schedule, and approval notification. Ineligible sellers receive transparent rejection reasons along with personalized coaching recommendations generated by the AI agent. The system also schedules automatic re-evaluation to reassess eligibility when business performance improves.

7. AI/ML Pipeline Diagram



Description

The AI/ML Pipeline Diagram presents the complete Machine Learning lifecycle implemented within the Meesho Pragati Agent. It explains how raw seller data is transformed into accurate, explainable, and production-ready lending predictions.

The pipeline begins with data ingestion from multiple business sources including orders, returns, ratings, dispatch records, advertising performance, repayment history, and seller profiles. During preprocessing, missing values are handled, outliers are removed, categorical variables are encoded, numerical attributes are normalized, and new behavioral features are engineered.

Exploratory Data Analysis is performed to understand feature distributions, correlations, and class imbalance before model development.

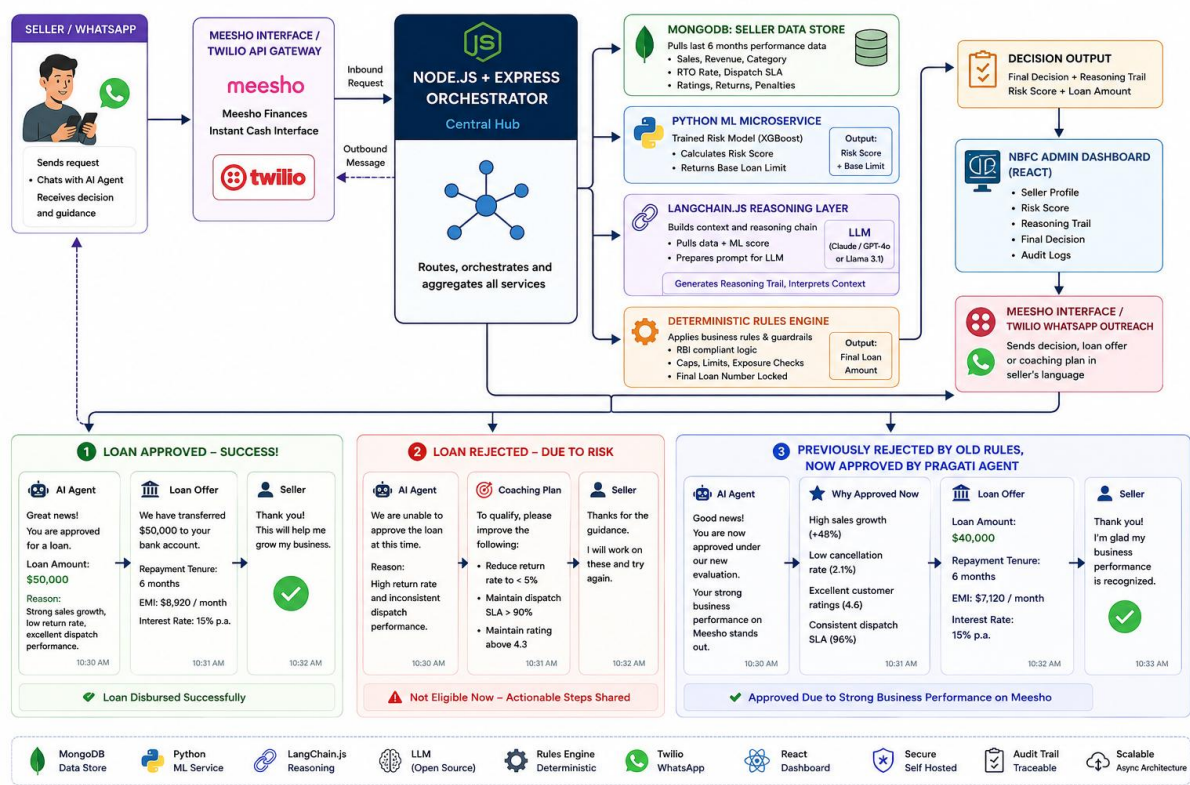
The core predictive model utilizes the **XGBoost (Extreme Gradient Boosting)** algorithm, selected for its superior performance on structured tabular financial datasets. XGBoost constructs an ensemble of gradient-boosted decision trees that iteratively minimize prediction errors while capturing complex nonlinear relationships between seller behavior and repayment risk.

Model validation is performed using train-validation-test splits together with K-Fold Cross Validation. Performance is evaluated using ROC-AUC, Precision, Recall, F1-Score, Accuracy, Log Loss, and Confusion Matrix.

To improve explainability, SHAP (SHapley Additive Explanations) is employed to calculate feature importance for every prediction. These explanations are supplied to the Agentic AI layer, enabling transparent decision explanations for both sellers and auditors.

Finally, the predicted risk score, suggested loan amount, confidence score, and SHAP explanations are forwarded to the deterministic Business Rules Engine, which generates the final RBI-compliant lending decision.

8. User Journey Diagram



Description

The User Journey Diagram illustrates the complete experience of a seller interacting with the Meesho Pragati Agent throughout the loan lifecycle. It captures both user actions and system responses at every stage of engagement.

The journey starts when the seller discovers the AI lending assistant through WhatsApp or the Meesho platform. After providing consent for data access, the system automatically retrieves historical business performance information and evaluates loan eligibility.

Following AI analysis and Machine Learning risk assessment, the seller receives a transparent lending decision along with detailed explanations and personalized coaching recommendations. If approved, the seller is guided through loan acceptance, disbursement, repayment scheduling, and post-loan support.

If rejected, the seller receives actionable insights explaining the reasons for rejection and specific recommendations for improving future eligibility. Continuous monitoring ensures that sellers are automatically notified when they become eligible for financing again, creating a long-term relationship rather than a one-time lending transaction.